

Automated Consumer Complaint Classification System

An NLP pipeline for routing financial complaints — from baseline to transformers

Dataset	CFPB Consumer Complaint Database
Usable records	199,678 complaints with narrative text
Target	8 product categories
Models	TF-IDF + Logistic Regression · DistilBERT · BERT
Hardware	NVIDIA RTX 5080 (16 GB) · CUDA 12.8
Date	June 08, 2026

Executive Summary

This project builds an AI system that automatically reads a free-text consumer complaint and classifies it into the correct financial-product category, enabling instant routing to the right department. Using **199,678** real complaints from the U.S. Consumer Financial Protection Bureau (CFPB), we trained and compared three models of increasing sophistication. The best model reached **86.8%** test accuracy across **8** categories, turning a slow, manual triage process into a sub-second automated decision.

Key outcomes

- **Baseline (TF-IDF + Logistic Regression):** 86.8% accuracy, weighted F1 0.8676 — fast, interpretable, near-zero inference cost.

Business value

- Cuts complaint triage from minutes of human reading to milliseconds of automated routing.
- Reduces mis-routing, which is a primary driver of slow resolution and repeat contacts.
- Surfaces category and sentiment trends that flag emerging product issues early.
- A tiered design (cheap baseline + accurate transformer) lets the business trade cost vs. accuracy.

1. Business Problem

Large financial institutions receive thousands of complaints every day through web forms, mobile apps, email, call centers, and social media. When these arrive as unstructured free text, a human must read each one and decide which team should handle it — credit cards, mortgages, debt collection, and so on. Manual review is slow, inconsistent, and expensive, and mis-routing directly worsens response time and customer experience.

Goal: build an AI system that, given the complaint narrative, automatically (1) classifies the complaint into its product category, (2) routes it to the correct department, and (3) exposes trends and sentiment for operational monitoring.

Worked example

Complaint text	Account charged twice for my credit card payment and customer support has not responded
Predicted category	Credit card or prepaid card
Routed to	Card Services team

2. Dataset Overview

The source is the public **CFPB Consumer Complaint Database**. The raw file contains **903,983** complaints, but only those where the consumer consented to publish their narrative contain text we can classify on — **199,970 rows (22.12%)**. After consolidating product labels and dropping empties, the modeling dataset is **199,678** complaints spanning **2015-03-19** to **2017-11-04**, across **3,299** companies and **62** states/territories.

Key fields used

Field	Role
Consumer complaint narrative	Input text — what the model reads
Product	Target label (consolidated to 8 categories)
Date received	Trend / volume analysis
Company, State	Operational segmentation & geography
Timely response?, Consumer disputed?	Downstream quality signals

Category consolidation

CFPB renamed and merged product taxonomies over the years, producing 18 raw labels that include exact duplicates (e.g. “Credit reporting” vs “Credit reporting, credit repair services, or other personal consumer reports”). We mapped these to **8 canonical business categories**, which removes label noise and improves model accuracy. Final class distribution:

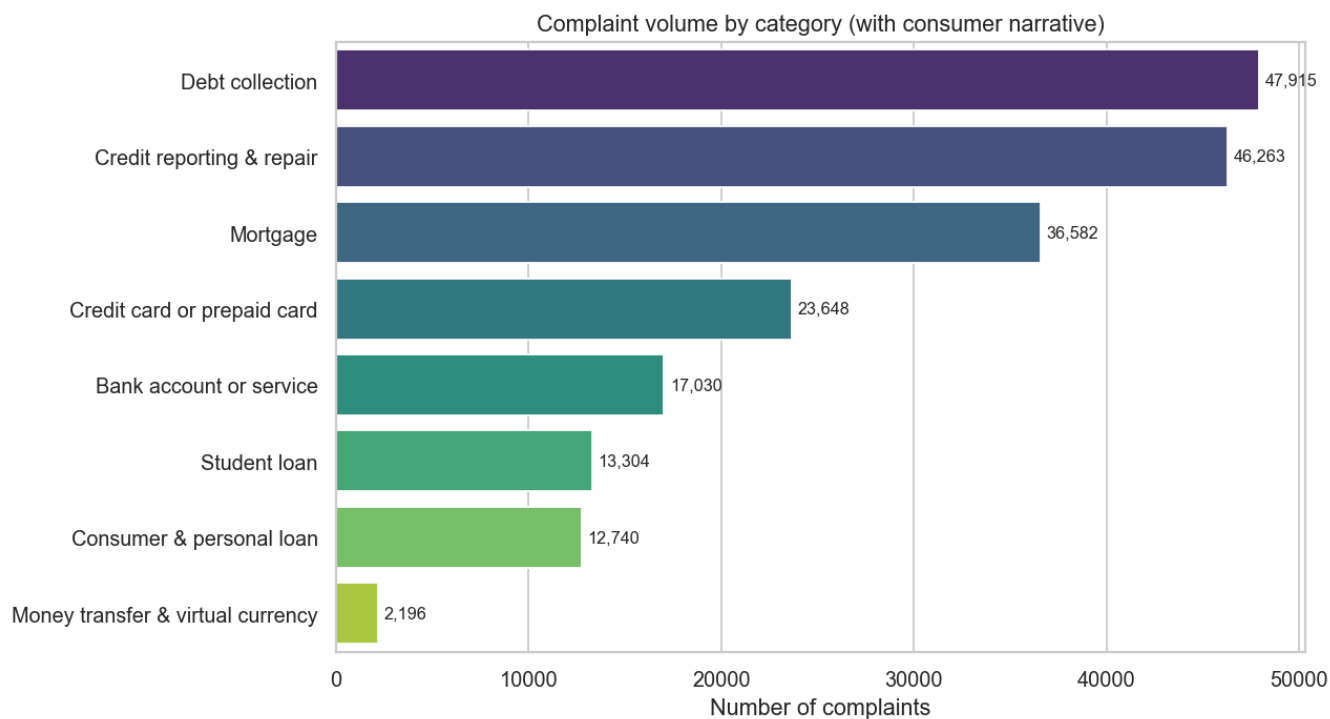
Category	Complaints	Share
Debt collection	47,915	24.0%
Credit reporting & repair	46,263	23.2%

Mortgage	36,582	18.3%
Credit card or prepaid card	23,648	11.8%
Bank account or service	17,030	8.5%
Student loan	13,304	6.7%
Consumer & personal loan	12,740	6.4%
Money transfer & virtual currency	2,196	1.1%

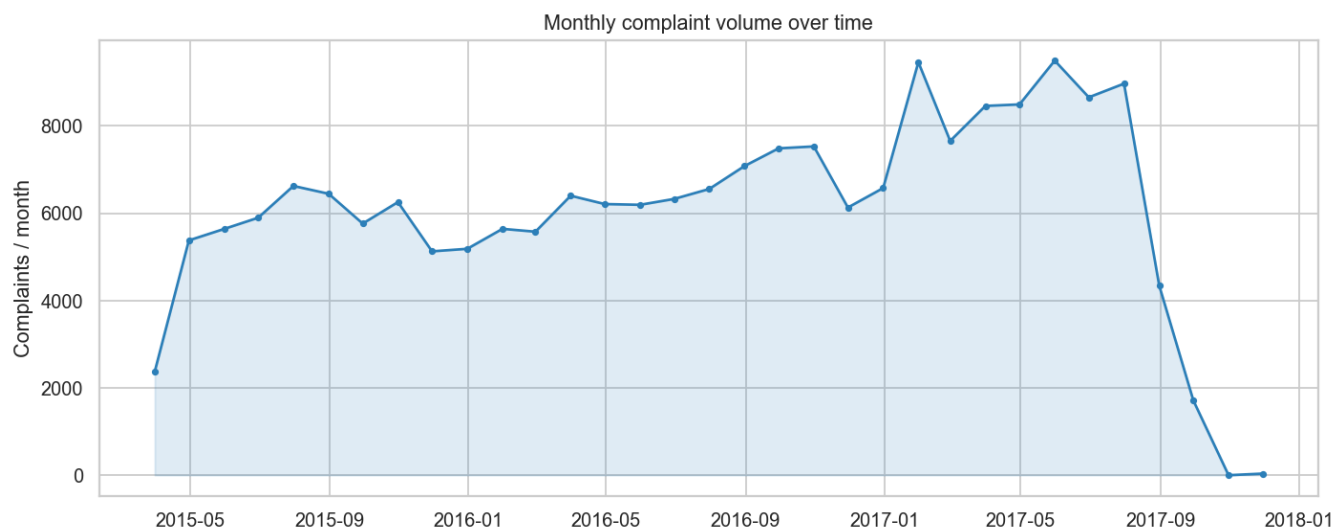
This is a genuine class-imbalance problem (the largest class is $\sim 20\times$ the smallest), which we track with macro-F1 in addition to accuracy.

3. Exploratory Data Analysis

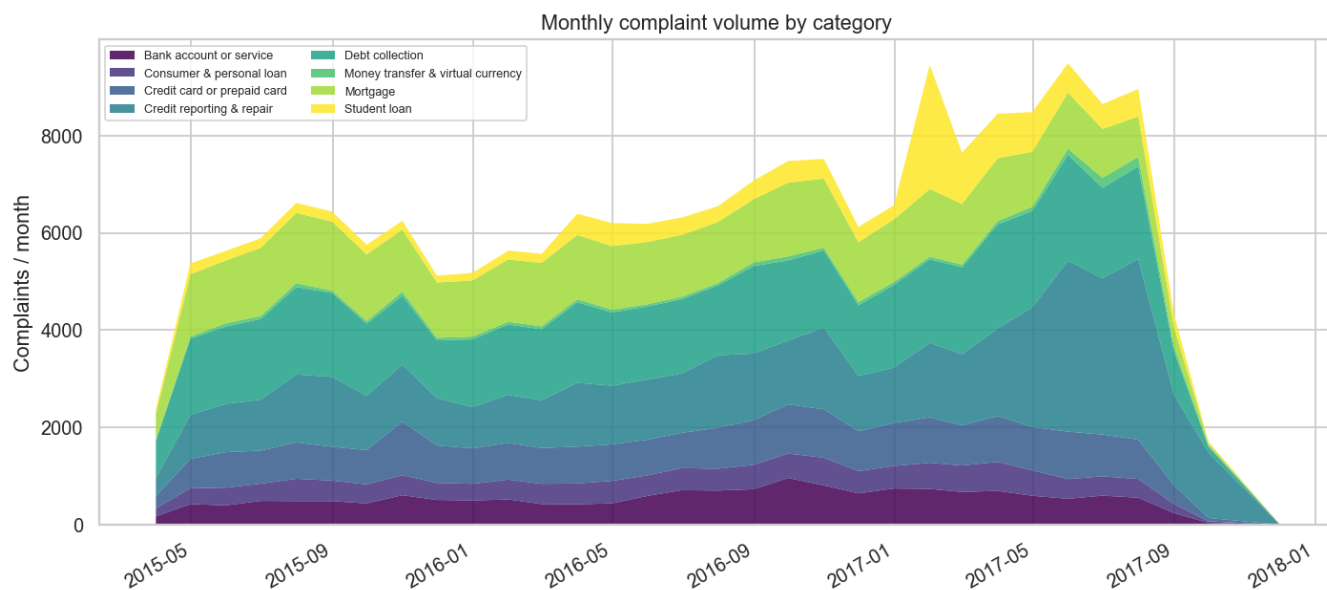
Before modeling, we profiled the data to understand volume, geography, text length, and vocabulary — each yielding an operational insight.



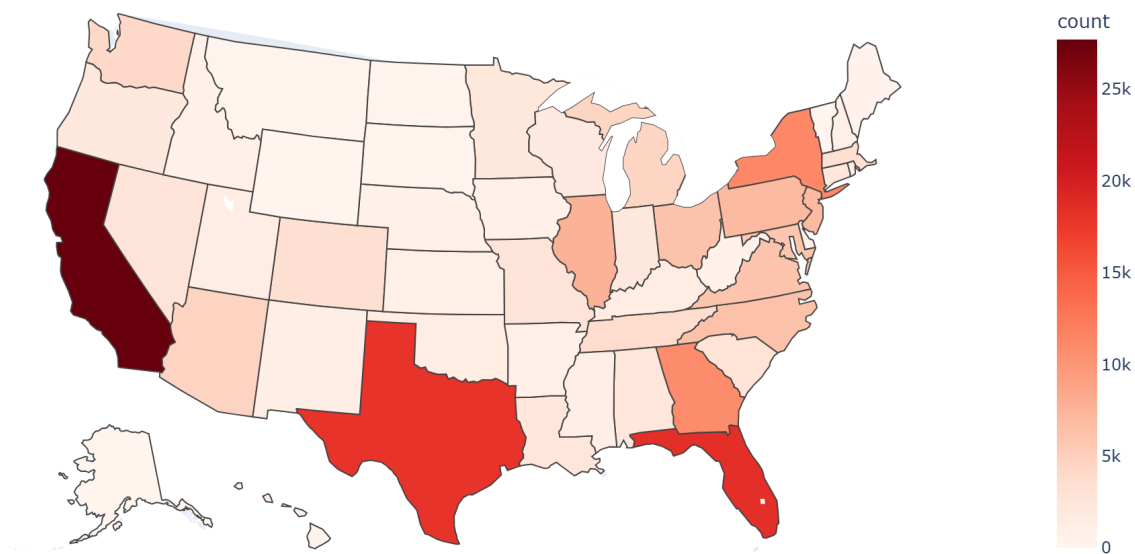
Debt collection and Credit reporting dominate the complaint mix — the two teams that would benefit most from automation.



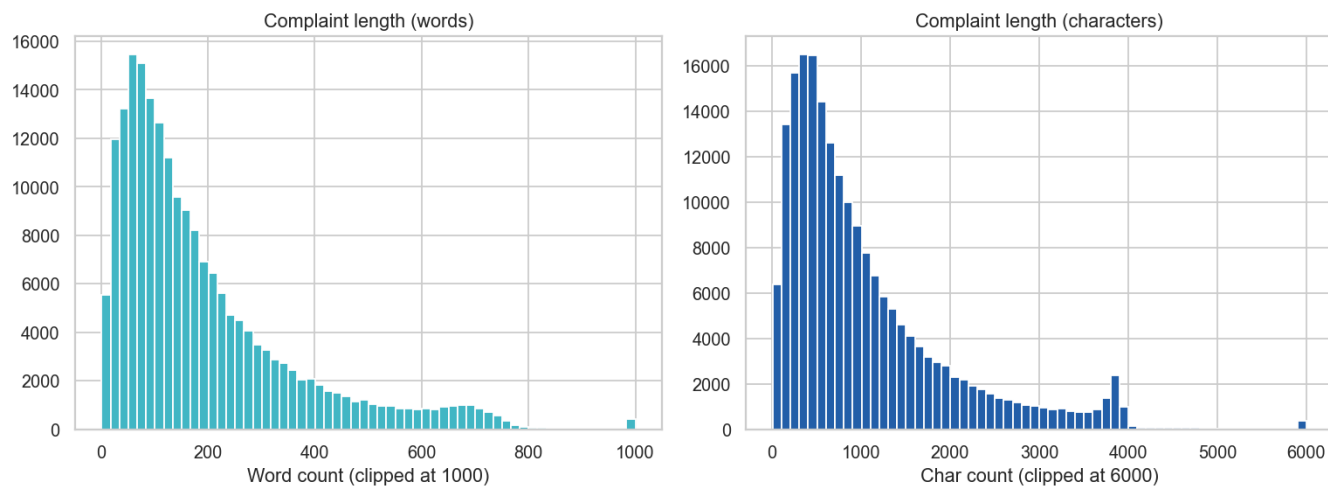
Monthly volume trends upward, peaking at 9,496 complaints in 2017-05-31 — staffing must scale with this curve.



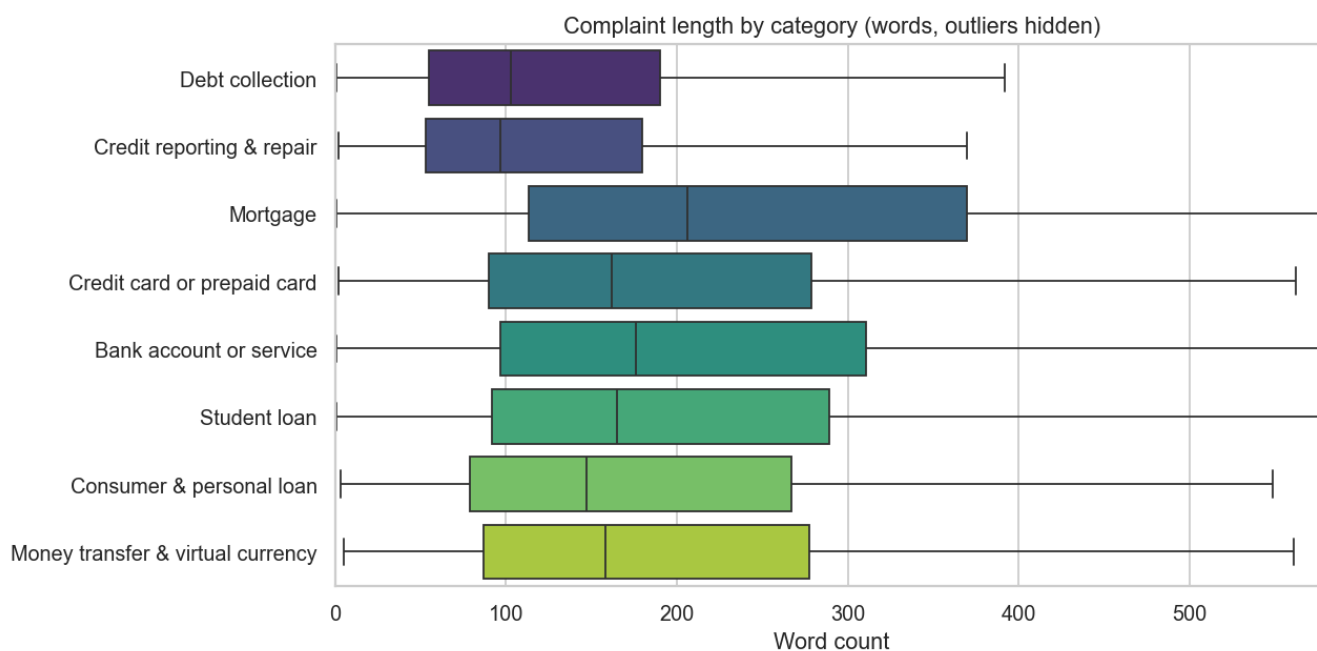
Complaint volume by U.S. state



Complaints concentrate in California, Texas, Florida, New York and Georgia — the large-population states.

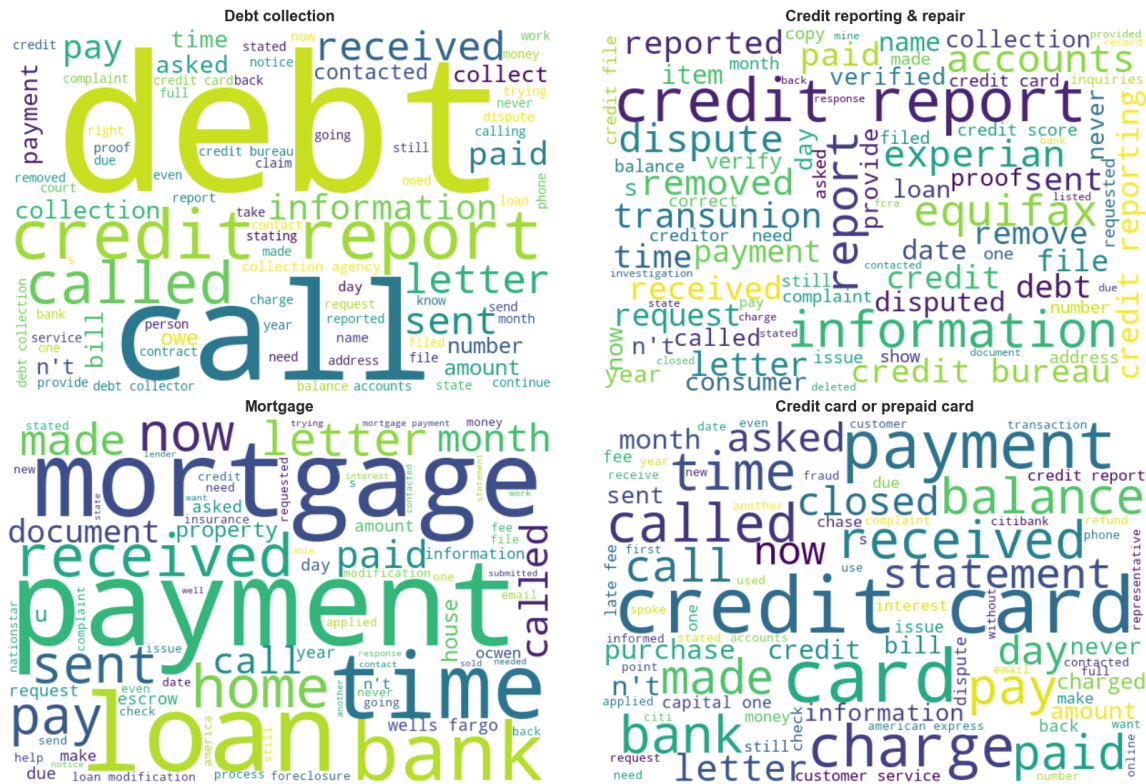


Most complaints are 100–300 words (median 136), but a long tail of detailed cases exists — informing a 256-token model limit.



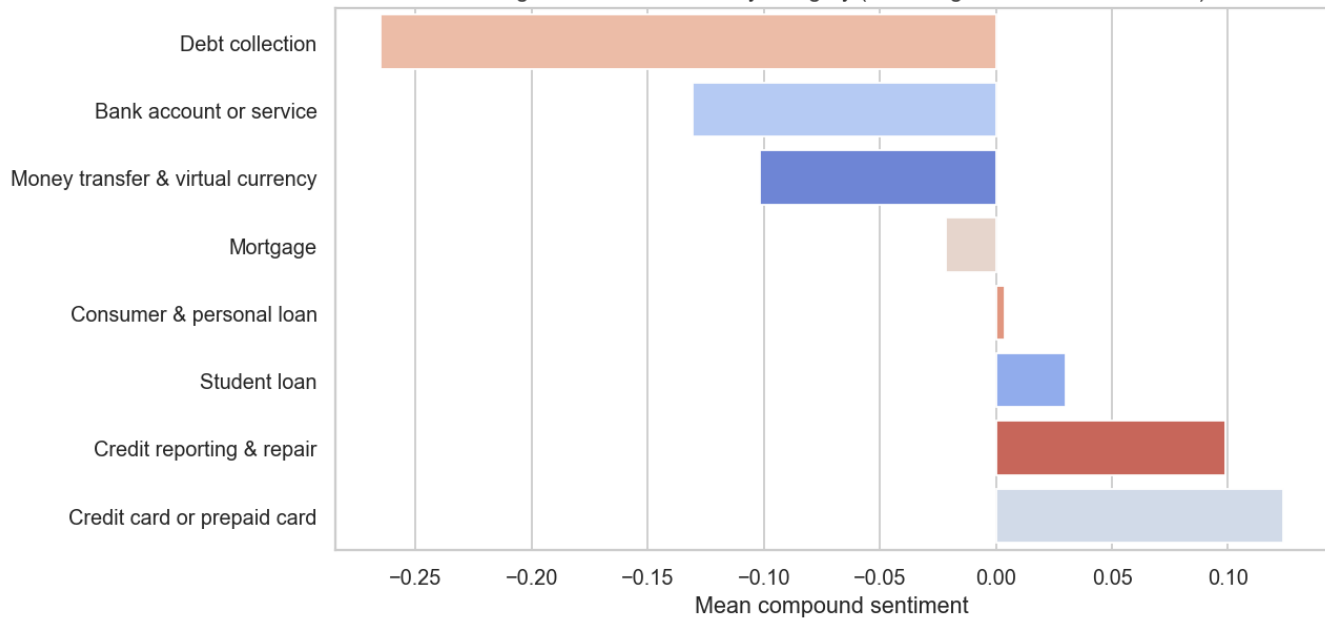
Mortgage and bank complaints run longest (more complex cases); debt-collection complaints are shortest.

Most frequent terms by complaint category



Category vocabularies are highly distinct — 'debt/credit/call', 'mortgage/payment/home', 'card/charge/payment' — which is exactly why text classification works well.

Average VADER sentiment by category (more negative = more distressed)



VADER sentiment is most negative for debt collection and bank complaints — useful as an urgency/priority signal.

4. Preprocessing & Feature Engineering

Text cleaning pipeline

For the classical model, each narrative passes through a standard NLP cleaning pipeline. (Transformers consume raw text and handle tokenization internally.)

- Lowercase and strip HTML / URLs
- Remove CFPB PII-redaction blocks (runs of 'XXXX')
- Remove punctuation and non-alphabetic characters
- Remove English stop-words and very short tokens
- Lemmatize each token to its dictionary root (WordNet)

Before / after example

Before	Report that I have previously disputed that has yet to be removed this information is more then seven years old meet credit reporting requirement...
After	credit report previously disputed yet removed information seven year old meet credit reporting requirement...

Engineered numeric features

Beyond the text, we engineered signals that capture complaint complexity and customer frustration — useful for urgency scoring and as auxiliary insight:

Category	Words	Chars	Sent.	CAPS	Excl.	Sentiment
Bank account or service	230.8	1248.4	12.8	15.6	0.4	-0.13
Consumer & personal loan	202.2	1079.5	11.4	13.2	0.3	+0.00
Credit card or prepaid card	210.9	1143.5	11.8	13.1	0.3	+0.12
Credit reporting & repair	141.1	787.0	7.3	15.3	0.2	+0.10
Debt collection	149.9	814.8	8.1	11.3	0.2	-0.27
Money transfer & virtual currency	211.2	1142.3	11.7	15.3	0.3	-0.10
Mortgage	267.2	1470.1	14.6	21.0	0.4	-0.02
Student loan	217.9	1182.7	12.0	11.6	0.3	+0.03

Interpretation: longer, more capitalized, more negative complaints (mortgage, debt collection) tend to be higher-stakes cases worth prioritizing.

5. Modeling Approach

We trained three models of increasing capacity on an identical stratified 80/20 split (159,742 train / 39,935 test), so every result is directly comparable.

Model	What it is	Why include it
TF-IDF + Logistic Regression	Bag-of-words (1–2 grams) + linear classifier	Fast, interpretable, cheap baseline
DistilBERT	Distilled transformer (66M params)	10% smaller/faster, near-BERT accuracy — production
BERT-base	Full transformer (110M params)	Strongest contextual understanding

6. Hyperparameter Tuning

Baseline — 5-fold grid search

We tuned the inverse-regularization strength **C** of the logistic-regression head with 5-fold cross-validation (scoring weighted-F1), over a multinomial lbfgs/L2 configuration.

C	Mean CV weighted-F1	Std
2.0	0.8666	0.0004
5.0	0.8661	0.0013
1.0	0.8638	0.0007
10.0	0.8635	0.0010
0.5	0.8586	0.0009
0.1	0.8340	0.0019

Best: C = 2.0 (CV weighted-F1 0.8666). Tuning ran in 277.9s across 30 fits.

Transformers — training configuration

Transformers were fine-tuned with the AdamW optimizer and a linear warmup schedule. Key hyperparameters:

Hyperparameter	DistilBERT	BERT
Learning rate	—	—
Batch size	—	—
Epochs	—	—
Max sequence length	—	—
Weight decay	—	—
Warmup ratio	—	—
Mixed precision	—	—

7. Results & Model Comparison

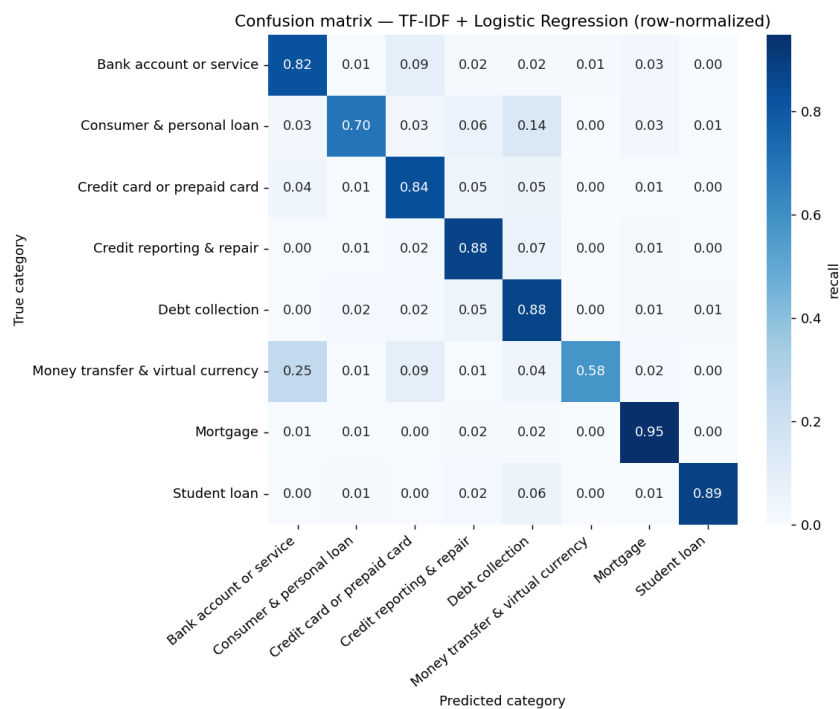
Metric	TF-IDF + LogReg
Accuracy	0.8684
Weighted F1	0.8676
Macro F1	0.8335
Weighted Precision	0.8678
Weighted Recall	0.8684

Higher is better on every metric. Macro-F1 (unweighted class average) is the honest measure under class imbalance.

Per-category F1

Category	TF-IDF + LogReg
Bank account or service	0.831
Consumer & personal loan	0.747
Credit card or prepaid card	0.832
Credit reporting & repair	0.881
Debt collection	0.859
Money transfer & virtual currency	0.669
Mortgage	0.941
Student loan	0.907

Confusion matrices (best models)



TF-IDF + Logistic Regression — strong diagonal; most confusion is between the two credit-related categories, which overlap semantically.

8. Business Value & Recommendations

Operational impact

- **Speed:** routing decisions drop from minutes of human reading to a few milliseconds per complaint.
- **Accuracy:** consistent, policy-aligned categorization removes human variance and mis-routing.
- **Cost:** the baseline model runs on CPU at negligible cost; transformers add accuracy when it matters.
- **Scalability:** the same model absorbs volume spikes (like the observed peak) with no extra headcount.
- **Insight:** category, geography, and sentiment trends become a live early-warning dashboard.

Recommended deployment

Adopt a **tiered routing strategy**: run the cheap TF-IDF baseline on every complaint; when its confidence is low, escalate to the fine-tuned transformer for a second opinion. This captures most of the transformer's accuracy at a fraction of the compute cost. Pair the category prediction with the engineered sentiment/urgency features to prioritize the most distressed customers first.

Next steps

- Add a confidence threshold + human-in-the-loop review for low-certainty predictions.
- Address class imbalance (class-weighted loss / focal loss) to lift minority-category recall.
- Extend beyond category to predicted urgency and automated complaint summarization.
- Monitor for data drift as products and complaint language evolve over time.

Reproducibility

Full pipeline: `data_prep` → `eda` → `features` → `train_baseline` → `train_transformer` → `report_gen`.
All code in `src/`, figures in `outputs/figures/`, metrics in `outputs/metrics/`.